

Cyber Risk & Insurance

Background Guide for Agentic AI Development

3-Week Internship Project
June 2026

Table of Contents

1. Introduction to Cyber Risk
2. Understanding Cyber Risk Severity: CVSS Scores
3. The Cost of Data Breaches (2025-2026)
4. Cyber Insurance Market Overview
5. Common Cyber Attack Vectors
6. Industry-Specific Risk Profiles
7. Cyber Insurance Underwriting Factors
8. References

1. Introduction to Cyber Risk

Why Cyber Insurance?

Cyber insurance is one of the fastest-growing insurance segments globally. Unlike traditional property or liability insurance, cyber risk is unique: it evolves rapidly, affects all industries, and the relationship between technical security controls and financial impact is non-linear. This makes it ideal for an agentic AI underwriting model.

Market Context (2025-2026)

Metric	2025 Value	2026 Projection
Global Cyber Insurance Market	\$15.6 billion	\$16.4 billion
US Average Breach Cost	N/A	\$10.22 million (all-time high)
Global Average Breach Cost	\$4.88 million	\$4.44 million (9% decline)
Expected Annual Growth Rate	15-20%	15-20% through 2030
Projected Market Size (2030)	N/A	\$40+ billion

Key Insight: The global average fell due to better AI-driven detection, but the US average hit a record high due to regulatory penalties and litigation costs. This divergence is crucial for underwriting models. [1][2]

2. Understanding Cyber Risk Severity: CVSS Scores

The Common Vulnerability Scoring System (CVSS) is the industry standard for rating software vulnerability severity. All actuarial cyber models reference CVSS as a foundational risk metric. [3][4]

CVSS Basics

Scale: 0.0 (minimal) to 10.0 (most severe)

Maintained by: FIRST (Forum of Incident Response and Security Teams)

Current Version: CVSS v4.0 (released 2024)

Database: National Vulnerability Database (NVD) tracks 200,000+ vulnerabilities [5][6]

Severity Ratings

Score Range	Severity	Interpretation
0.0	None	No vulnerability
0.1 - 3.9	Low	Unlikely to impact most systems
4.0 - 6.9	Medium	Impacts some systems; may require patches
7.0 - 8.9	High	Significant risk; patch urgently
9.0 - 10.0	Critical	Severe impact; patch immediately

What CVSS Does NOT Measure: While CVSS indicates technical severity, it does NOT measure actual risk in your specific environment. A critical vulnerability on an unexposed internal system carries less real risk than a medium-severity vulnerability on an internet-facing server. This is why cyber underwriting adds context. [7]

For Underwriting: Use CVSS as one input, not the whole story. Combine with company size, industry, exposure, and past breach history.

3. The Cost of Data Breaches (2025-2026)

Global Averages

Metric	2024	2025	Trend
Global Average	\$4.88M	\$4.44M	↓ 9% (AI adoption)
US Average	N/A	\$10.22M	↑ Record high
Healthcare (highest)	\$12.7M	\$11.2M	↓ Slight decline
Financial Services	N/A	\$6.08M	2nd highest cost
Time to Detect	258 days	241 days	↓ 17 days faster
Time to Contain	85 days	60 days	↓ 25 days faster

Industry Impact

Healthcare: Highest cost at \$11.2M per breach (2.5x global average). HIPAA fines, patient notification, and operational disruption drive severity. [8]

Financial Services: \$6.08M average. Regulatory penalties and reputation damage dominate. [9]

Critical Infrastructure: \$4.82M average but with disproportionate systemic risk.

Retail/Tech: Lower average costs but highest breach frequency.

Key Insight for Underwriting: The US divergence is critical. Organizations in the US face state-level privacy laws (California's CCPA, Virginia's VCDPA, etc.), SEC disclosure requirements, class-action lawsuits, and regulatory fines (FTC, state attorneys general). This explains why US breach costs are 2.3x the global average. [10][11]

Insider Threats (2026 Focus): Insider breaches average \$4.9B and are harder to detect. 95% of breaches involve human error at some stage. [12]

4. Cyber Insurance Market Overview

Market Growth & Projections

Current Market Size: ~\$15.6 billion (2025)

2026 Projection: \$16.4 billion (+5.1%)

Long-term Growth: 15-20% annually through 2030

2030 Target: \$40+ billion market [13][14][15]

Why Growth Is Explosive

1. **Regulatory Mandate:** Governments now require cyber disclosure. SEC rules push companies to carry cyber insurance.
2. **Underinsurance:** 88% of C-suite executives feel inadequately protected despite knowing the risk. [16]
3. **SME Expansion:** Small/mid-market businesses are adopting cyber insurance rapidly as rates drop and automation enables underwriting at scale. [17]
4. **Supply Chain Focus:** 70% of organizations experienced a third-party breach in 2025, driving demand for vendor risk insurance. [13]
5. **AI Uncertainty:** New AI-driven vulnerabilities are creating appetite for coverage.

2026 Market Dynamics

Capacity Expansion: Reinsurers deployed \$250M+ in new cyber capacity in H1 2025. This allows primary insurers to write more volume. [13]

Underwriting Discipline: After losses in 2023-2024, insurers are shifting from volume to profitability. Expect tighter underwriting criteria. [18]

Rate Stabilization: After 2025's 11% premium decrease, rates are expected to stabilize or slightly increase. [17]

Risk-Based Pricing: Companies with strong security postures receive better rates. Poor cyber hygiene = higher premiums or denial. [13]

Market Leaders (Buyers)

Large Enterprises (54% of market): \$10M+ annual spend, global policies, custom limits, incident response retainers. [17]

BFSI Sector (32% of market): Banks, insurance, financial services—highest risk, strictest underwriting. [17]

SME Segment (Fast-growing): Automated underwriting now enables quick quotes & lower premiums for small businesses. [17]

5. Common Cyber Attack Vectors

Leading Initial Access Methods (2025-2026)

Attack Vector	Frequency	Avg Breach Cost	Trend
Phishing (Email)	37% of ransomware	\$4.88M	Declining slightly
Stolen Credentials	22% of breaches	Varies	Rising
Software Vulnerability	20% of breaches	Varies	Now beating credentials
Voice Phishing (Vishing)	11% initial access	Varies	442% increase YoY
Social Engineering	25% overall	Varies	Shifting email→voice
Insider Threats	65% of breaches	\$4.9B per incident	48% increase 2026

Ransomware: The Dominant Threat for Underwriters

Ransomware is the #1 concern for CISOs and the primary driver of cyber insurance claims. [18]

- **Attack Frequency:** Every 11 seconds globally [19]
- **Industry Impact:** SMEs: 51% of costs; Financial: 9% YoY increase
- **Evolution:** Double extortion (steal + encrypt), targeting backups, insider recruitment [20]
- **Cost Trend:** Ransom payouts shrinking but business interruption costs rising

AI-Powered Attacks (Emerging 2025-2026)

- **AI Phishing:** 54% click rate vs 12% traditional (highly personalized emails) [21]
- **Deepfakes:** Video attacks up 2,137% over 3 years [19]
- **Model Evasion:** Attacks designed to evade AI-powered defenses [8]
- **Shadow AI Risk:** Unvetted AI tools add \$670K to breach costs [8]

Underwriting Implication: Companies without AI detection and those deploying AI without controls face higher risk. [22]

6. Industry-Specific Risk Profiles

Industry	Primary Threat	Avg Cost	Underwriting Factor
Healthcare	Ransomware, insiders	\$11.2M	HIPAA, patient data scale
Finance	Credential abuse	\$6.08M	Regulatory exposure
Public Admin	Geopolitical attacks	Varies	Critical infrastructure
Retail	Payment theft, POS	Lower	PCI-DSS compliance
Tech/SaaS	Zero-day, API abuse	Varies	Patch speed, deployment
Manufacturing	OT/IT convergence	Varies	Downtime costs

Why Industry Matters: Risk varies dramatically. Healthcare pays 2.5x more than average because patient data is high-value and regulatory penalties are severe. [8]

7. Cyber Insurance Underwriting Factors

Modern cyber underwriting balances technical controls with business context. Key assessment areas:

Organizational Risk Factors

- **Company Size:** Employee count (data exposure proxy)
- **Annual Revenue:** Larger = more data assets & higher impact
- **Industry:** Healthcare/Finance/Critical Infra = higher risk
- **Geographic Exposure:** US companies pay more (regulations); global = distributed risk
- **Customer Base:** B2C (customer data) vs B2B (lower exposure)

Technical Security Posture

- **Certifications:** SOC 2 Type II (gold), ISO 27001, NIST CSF
- **Security Tools:** SIEM, endpoint detection, AI security = 65% cost reduction [8]
- **Incident Response Plan:** Documented playbook + retainer = lower severity
- **Backup & Disaster Recovery:** Ransomware resilience (immutable backups)
- **Vulnerability Management:** Patch cadence, penetration testing frequency

Data & Compliance Risk

- **Data Classification:** PII, PHI, payment card data, trade secrets stored
- **Regulatory Compliance:** HIPAA, GDPR, CCPA, PCI-DSS, SOX adherence
- **Third-Party Risk:** Vendor concentration, SaaS dependencies
- **Past Breach History:** Prior incidents correlate with future risk or weak controls

Underwriting Decision Framework

HIGH RISK (Decline/strict controls): Healthcare without SOC 2 • Finance with poor patch management • Public infra with unpatched systems

MEDIUM-HIGH RISK: Retail with PCI-DSS + prior breach • SaaS with SOC 2 but no SIEM

MEDIUM RISK: Tech startup with ISO 27001 + incident plan • Manufacturer with reasonable posture

LOW RISK (Preferred): Enterprise with SOC 2 Type II + SIEM + AI • Global corp with comprehensive program

For Your Model: Combine company data (size, industry, certifications) with risk indicators (breach history, patch cadence, third-party exposure) to produce a risk score mapping to premium pricing and coverage terms.

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Note: All statistics and data points cited in this document are sourced from authoritative organizations including IBM Ponemon Institute, Verizon, Munich Re, Swiss Re, FIRST, the National Institute of Standards and Technology (NIST), and industry analysts. Where data was retrieved from aggregator sources (e.g., StationX, Bits From Bytes), the original research organizations (e.g., IBM, Ponemon, ITRC) are cited. Sources accessed June 2026.